

Machine Learning Answers (~20% Correct)

Q1:

Explain the concept of backpropagation in neural networks. (10 Marks)

Answer:

Backpropagation is used in neural networks to help the model learn from mistakes.

The network produces an output and compares it with the correct answer.

The error is then sent backward through the network to adjust weights.

This explanation does not clearly describe gradients or how updates are calculated.

Q2:

What is overfitting in machine learning? How can it be prevented? (5 Marks)

Answer:

Overfitting happens when a model learns too much from the training data.

It may perform well on training data but poorly on new data.

This answer does not explain detailed techniques to prevent it.

Q3:

Define gradient descent and explain its role in training neural networks. (5 Marks)

Answer:

Gradient descent is a method used to reduce error in machine learning models.

It changes the weights of the model step by step.

However, this explanation does not describe gradients or learning rate clearly.

Q4:

What is the difference between classification and regression in machine learning? (5 Marks)

Answer:

Classification and regression are types of machine learning problems.

Classification predicts categories while regression predicts numbers.

This answer does not include algorithm examples or detailed explanation.